

Neural Network

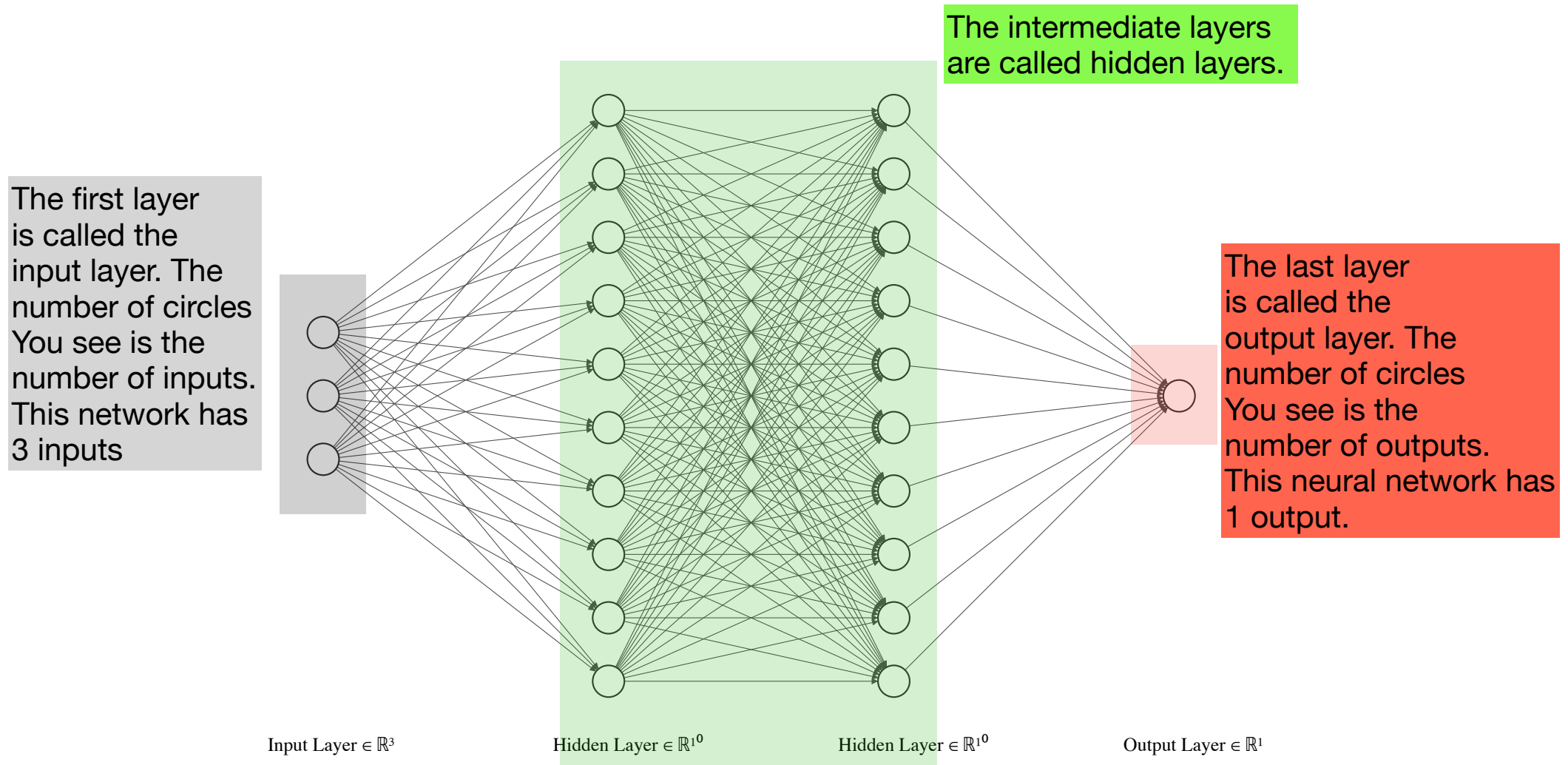
$$\vec{f}(\vec{x})$$

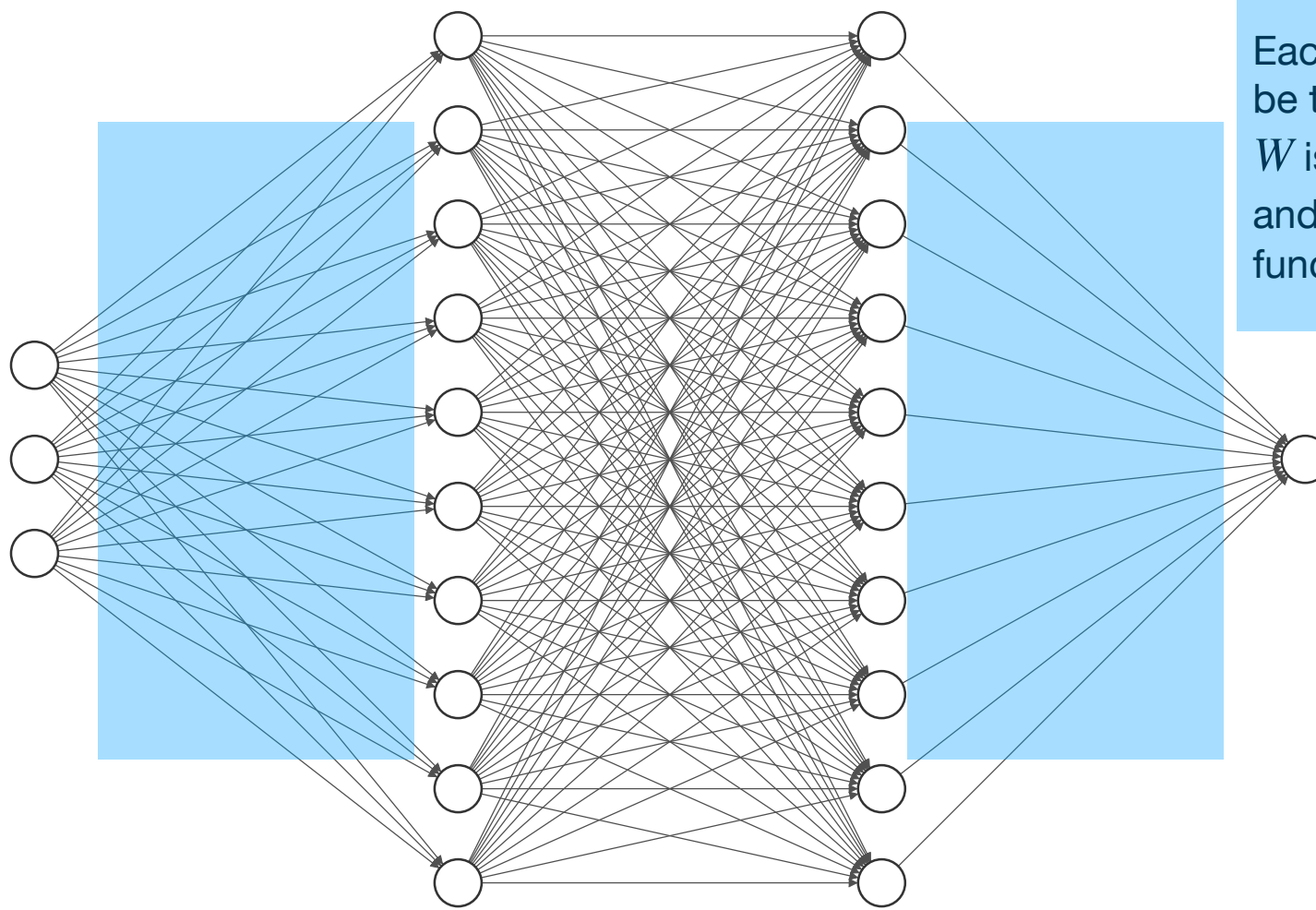
Neural Network are just very complex functions. In general, they can be written as $\vec{f}(\vec{x})$, $\vec{f}$ and $\vec{x}$ are vectors

$$\begin{aligned} f_1(x_1, x_2, \dots, x_n) \\ f_2(x_1, x_2, \dots, x_n) \\ f_3(x_1, x_2, \dots, x_n) \\ \vdots \\ f_m(x_1, x_2, \dots, x_n) \end{aligned}$$

You can think of the x_i as inputs and f_j as outputs. So the system of equations above have n inputs and m outputs.

Neural Network are can also be represented visually as





Input Layer $\in \mathbb{R}^3$

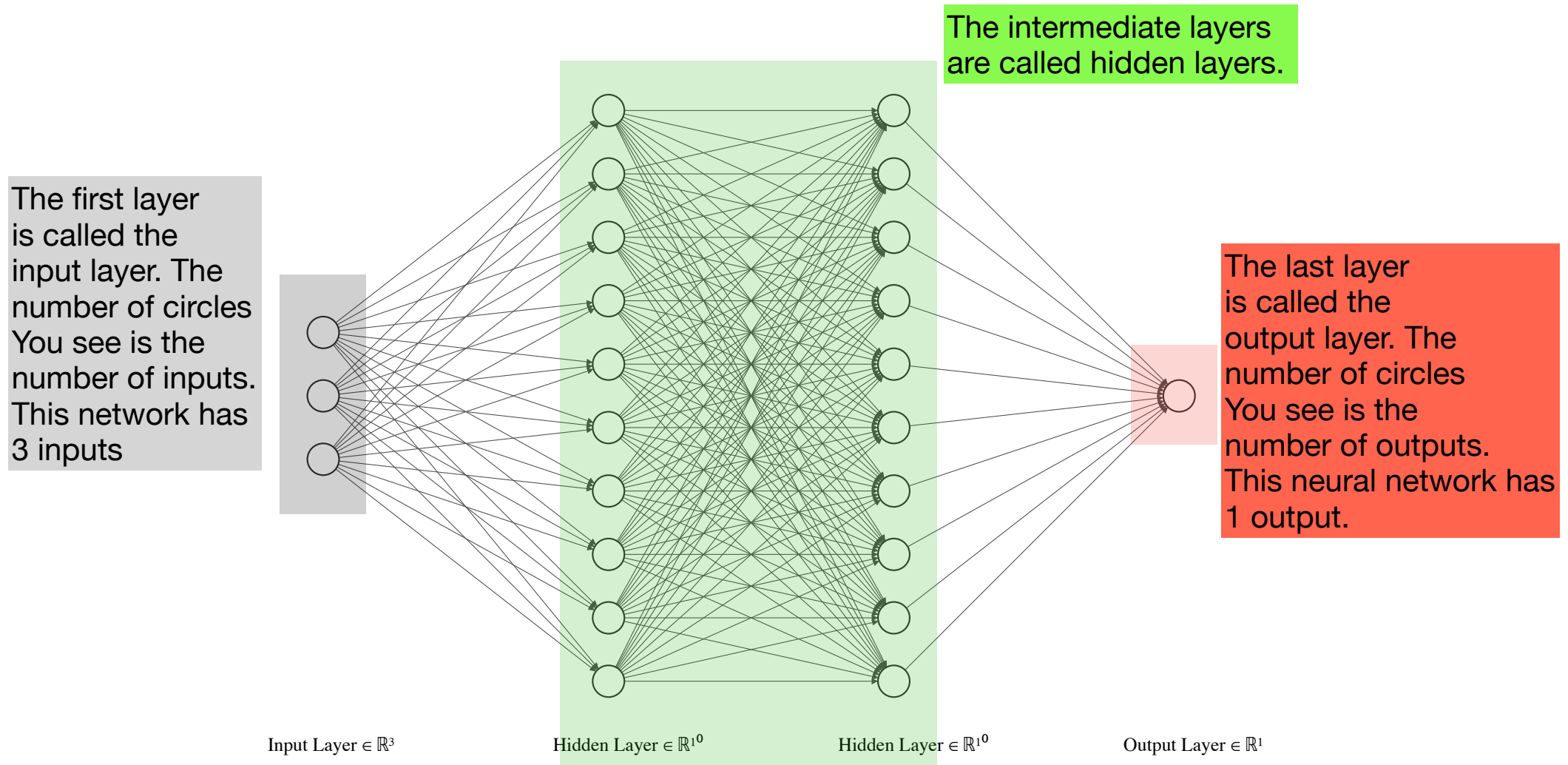
Hidden Layer $\in \mathbb{R}^{10}$

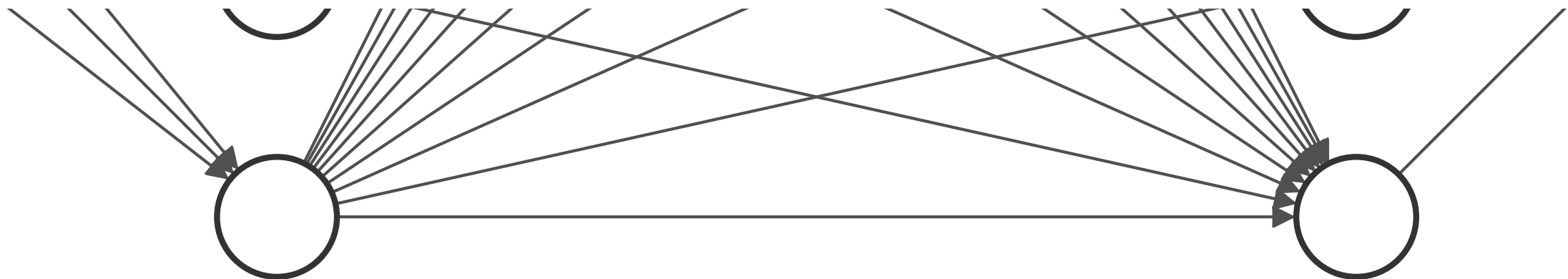
Hidden Layer $\in \mathbb{R}^{10}$

Output Layer $\in \mathbb{R}^1$

Each line that you see can be thought of as $\sigma(Wx + b)$.
 W is called the weights, b is the bias and $\sigma()$ is a nonlinear activation function.

Neural Network are can also be represented visually as





Hidden Layer $\in \mathbb{R}^{10}$

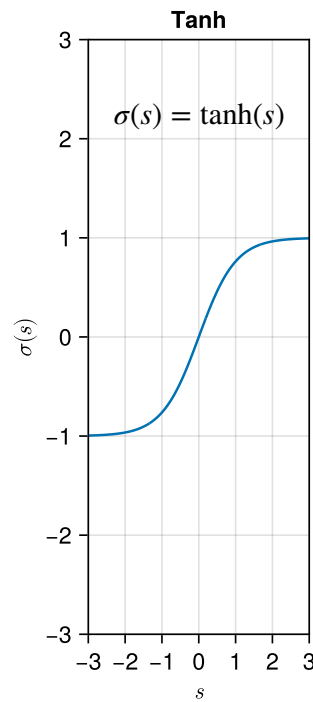
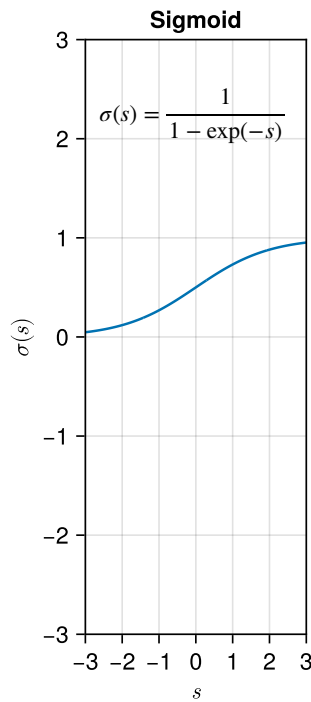
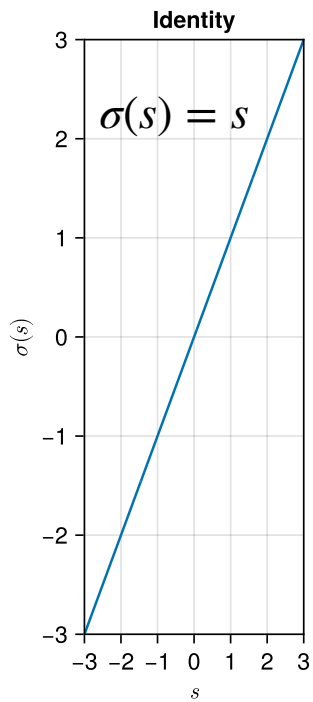
Hidden Layer $\in \mathbb{R}^{10}$



Input Layer $\in \mathbb{R}^1$

No hidden
layer(s)

Output Layer $\in \mathbb{R}^1$



$$NN(x) = y(x) = \sigma(Wx + b)$$

$$NN(x) = y(x) = Wx + b$$

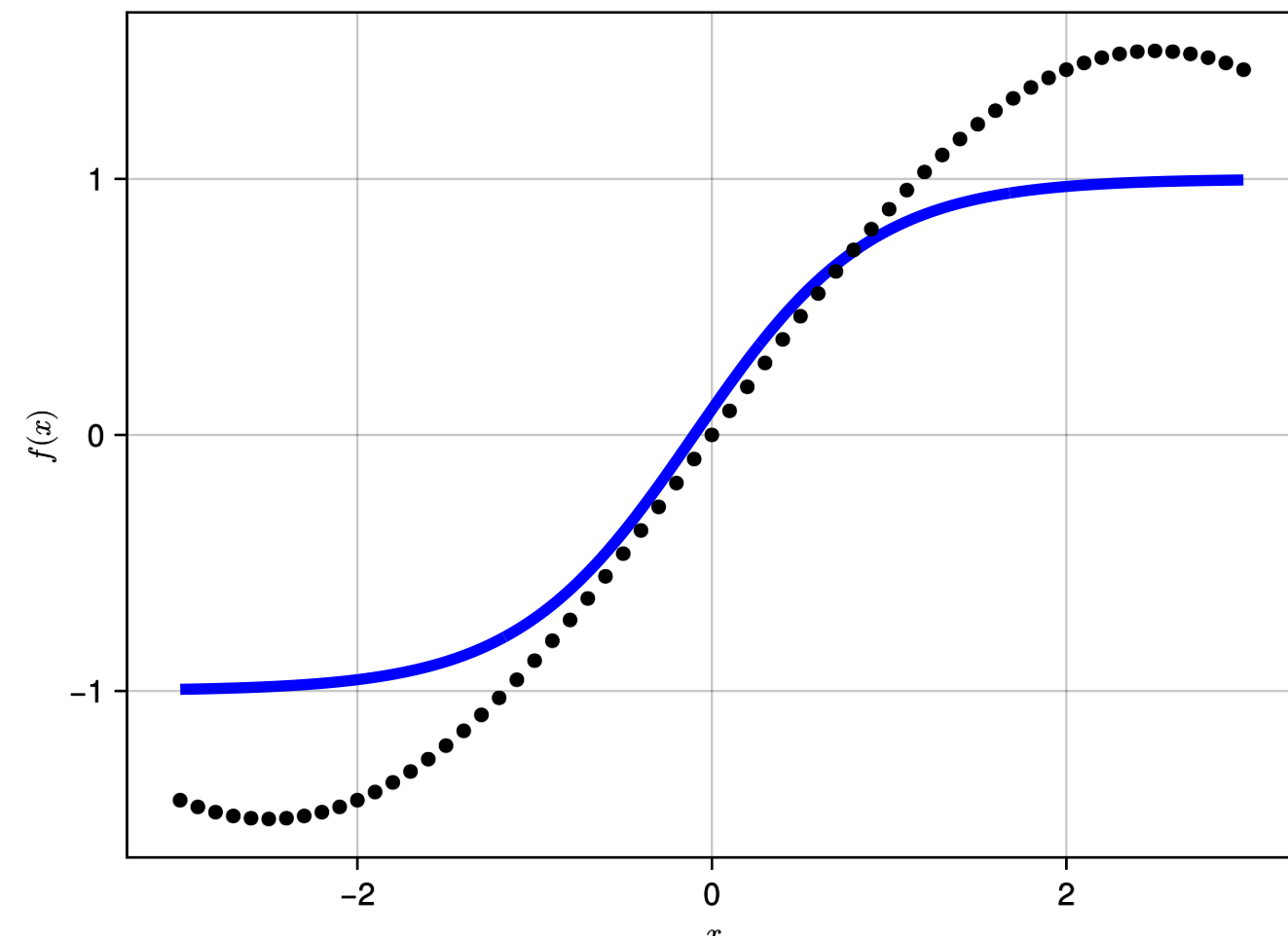
$$NN(x) = y(x) = \frac{1}{1 - \exp(-(Wx + b))}$$

$$NN(x) = y(x) = \tanh(Wx + b)$$

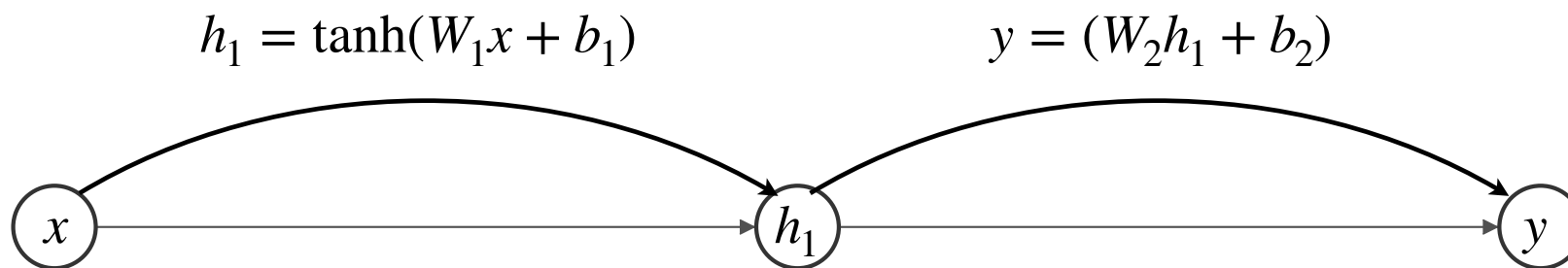


Input Layer $\in \mathbb{R}^1$

Output Layer $\in \mathbb{R}^1$



$$NN(x) = y(x) = \tanh(Wx + b)$$



Input Layer $\in \mathbb{R}^1$

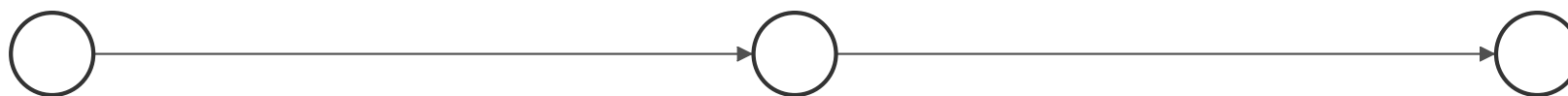
Hidden Layer $\in \mathbb{R}^1$

Output Layer $\in \mathbb{R}^1$

$$h_1(x) = \tanh(W_1 x + b_1)$$

$$y = W_2 h_1 + b_2$$

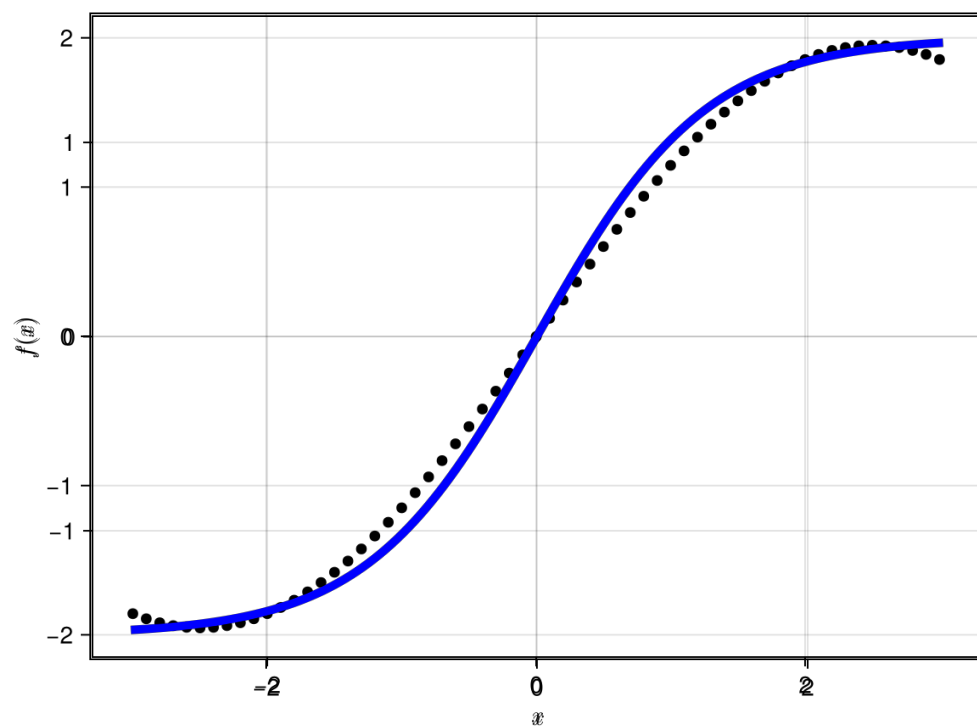
$$NN(x) = W_2 \tanh(W_1 x + b_1) + b_2$$



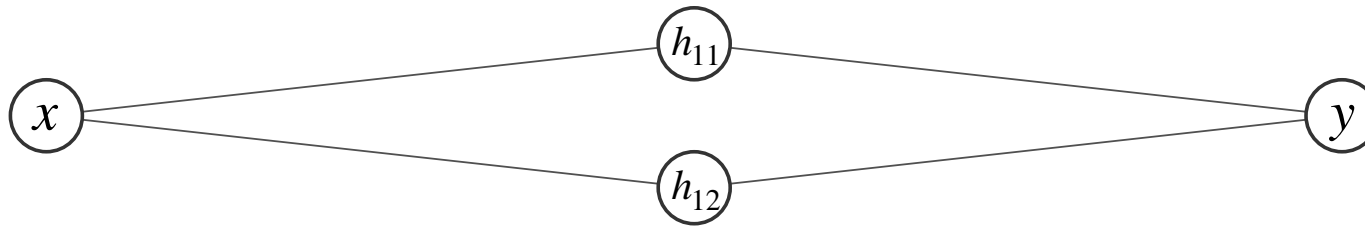
Input Layer $\in \mathbb{R}^1$

Hidden Layer $\in \mathbb{R}^1$

Output Layer $\in \mathbb{R}^1$



$$NN(x) = W_2 \tanh(W_1 x + b_1) + b_2$$



Input Layer $\in \mathbb{R}^1$

Hidden Layer $\in \mathbb{R}^2$

Output Layer $\in \mathbb{R}^1$

$$\begin{Bmatrix} h_{11} \\ h_{12} \end{Bmatrix} = \tanh \left(\begin{bmatrix} W_{11} \\ W_{12} \end{bmatrix} x + \begin{Bmatrix} b_{11} \\ b_{12} \end{Bmatrix} \right)$$

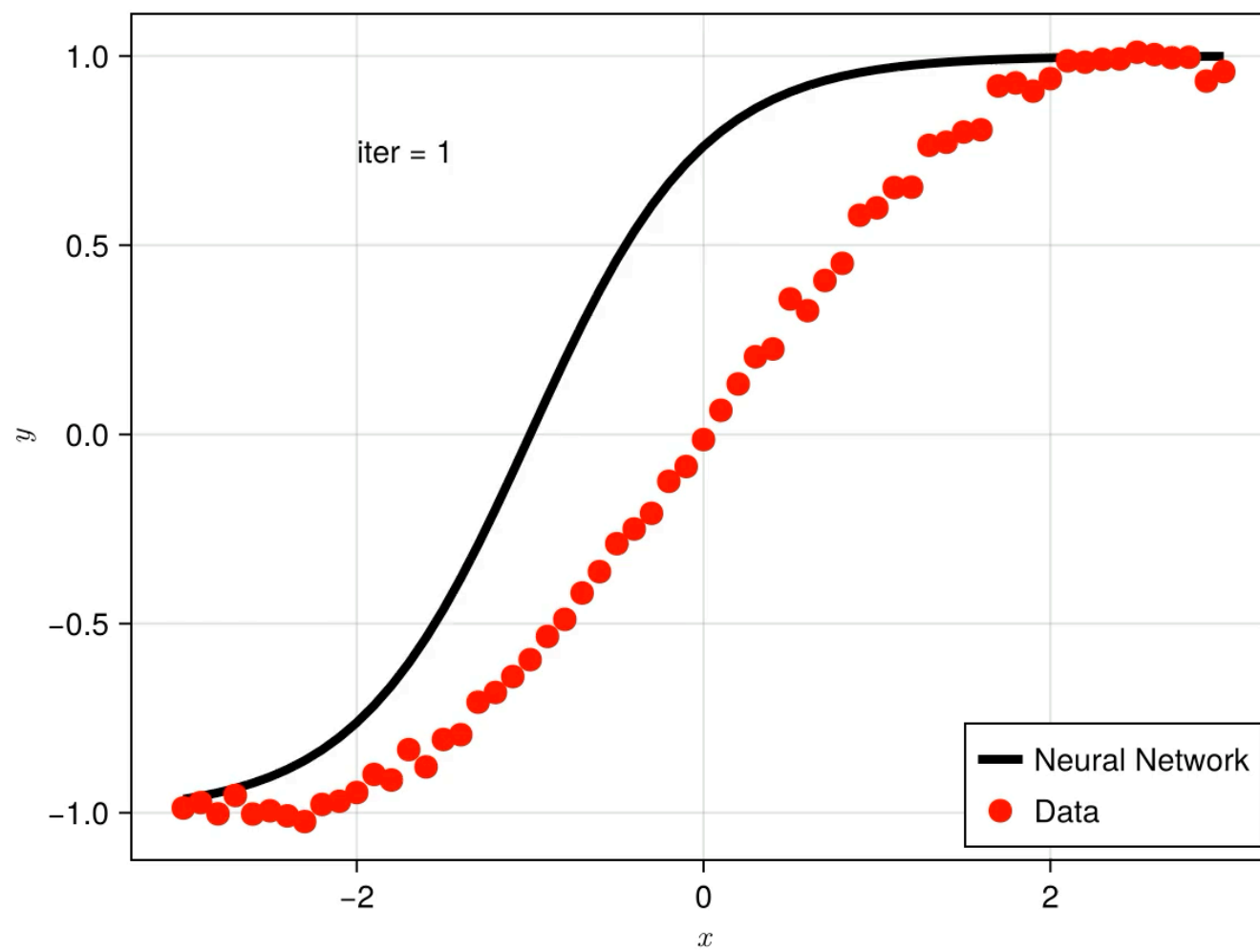
$$y(x) = NN(x) = \begin{bmatrix} W_{21} & W_{22} \end{bmatrix} \begin{Bmatrix} h_{11} \\ h_{12} \end{Bmatrix} + b_{21}$$

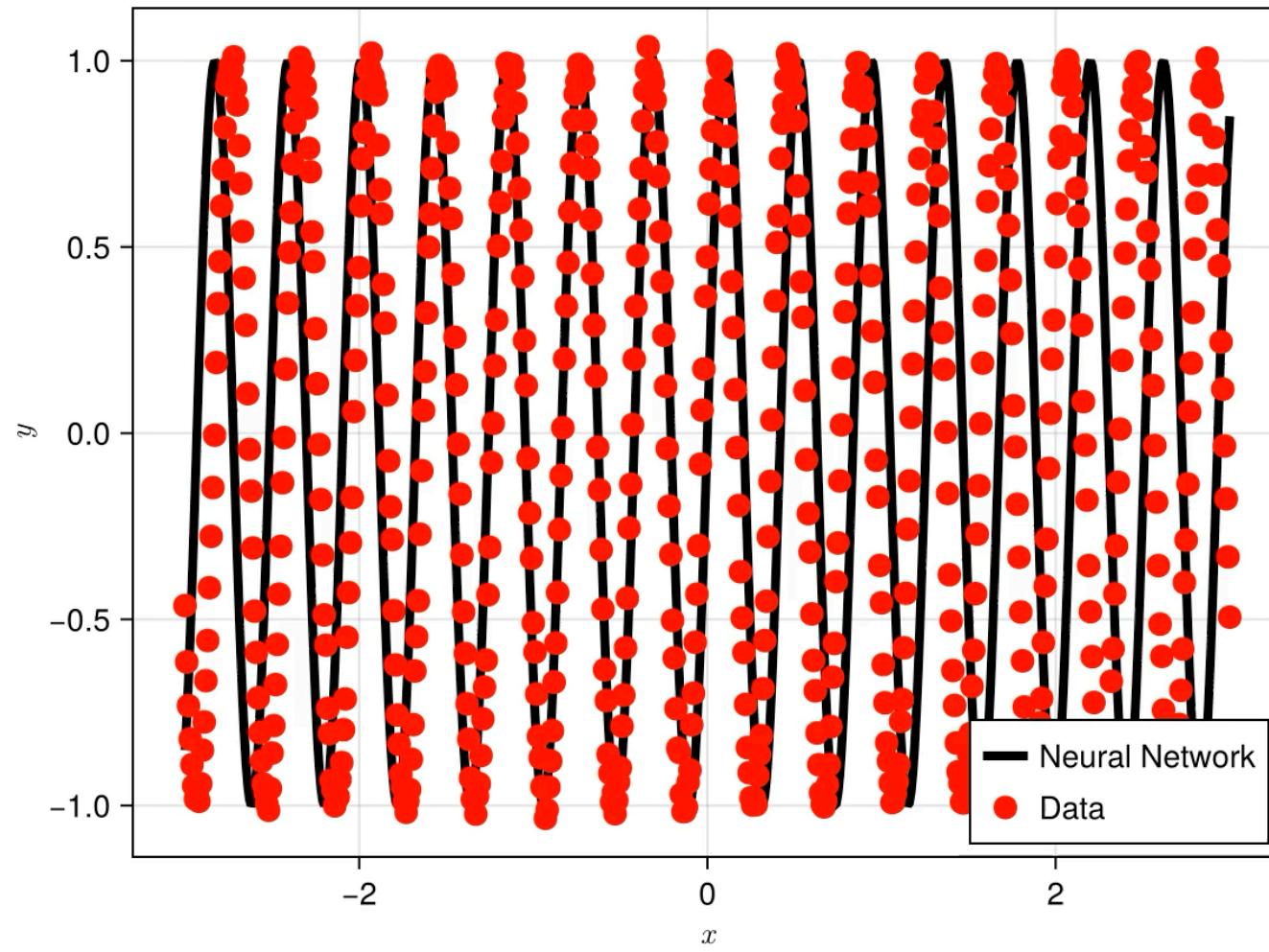
$$= \begin{bmatrix} W_{21} & W_{22} \end{bmatrix} \left\{ \tanh \left(\begin{bmatrix} W_{11} \\ W_{12} \end{bmatrix} x + \begin{Bmatrix} b_{11} \\ b_{12} \end{Bmatrix} \right) \right\} + b_{21}$$

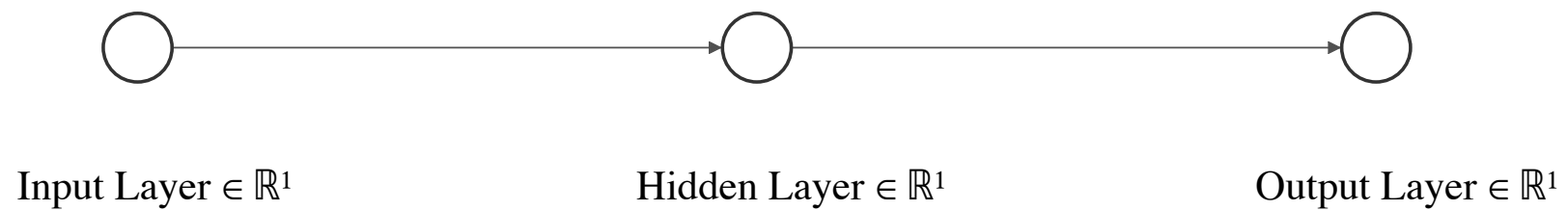
$$= \begin{bmatrix} W_{21} & W_{22} \end{bmatrix} \begin{Bmatrix} \tanh(W_{11}x + b_{11}) \\ \tanh(W_{12}x + b_{12}) \end{Bmatrix} + b_{21}$$

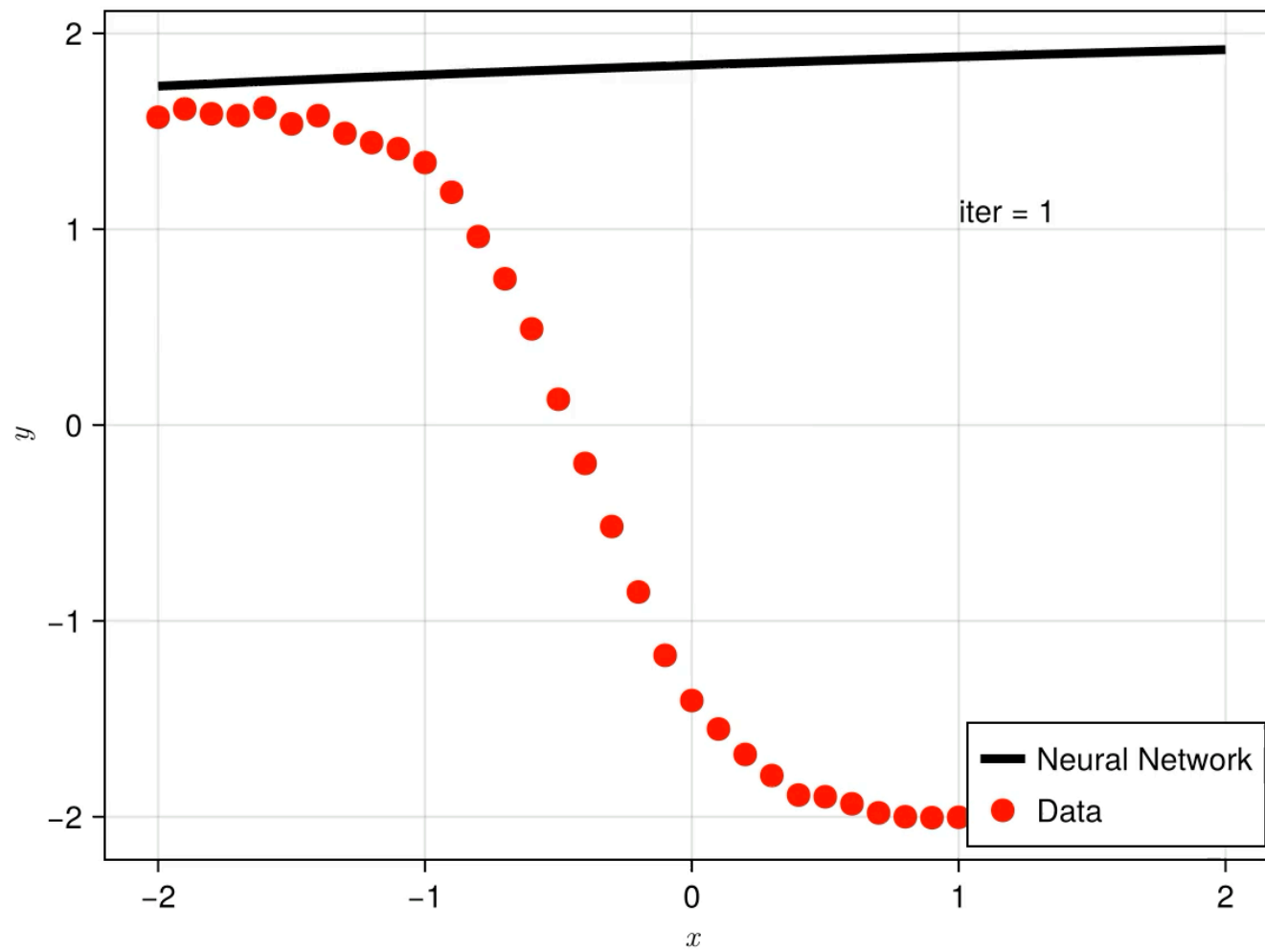
$$y(x) = NN(x) = W_{21} \tanh(W_{11}x + b_{11}) + W_{22} \tanh(W_{12}x + b_{12}) + b_{21}$$

“Data01.csv”

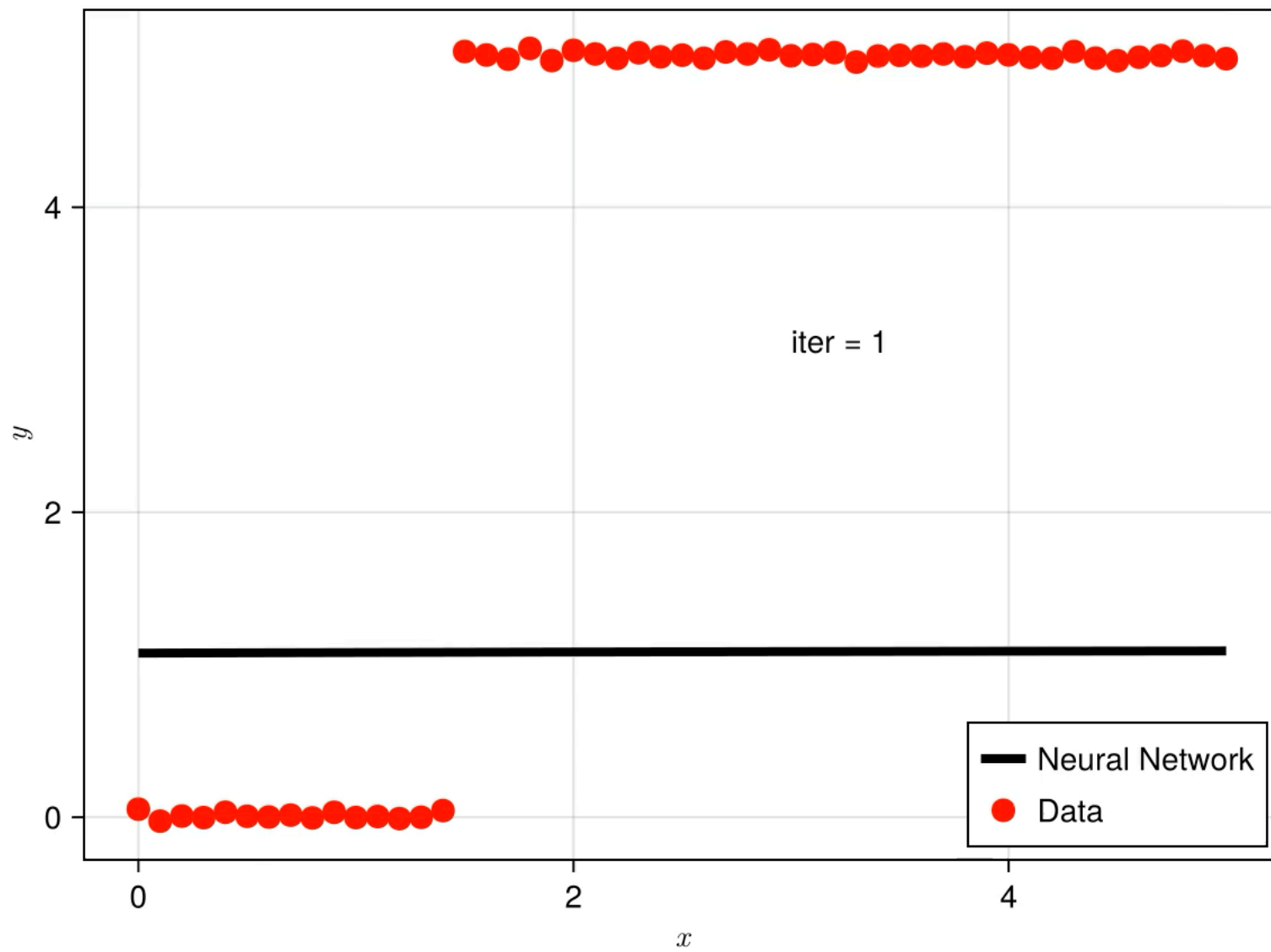




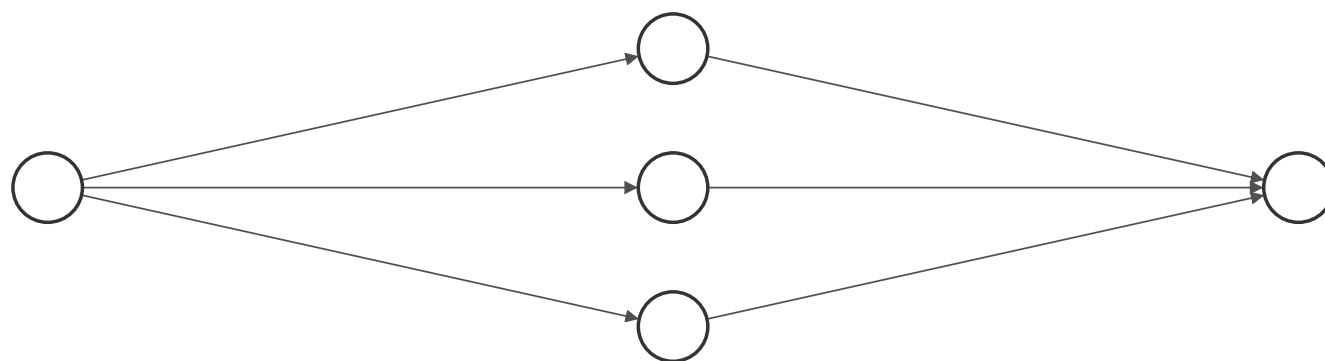




Data03.csv



Data03.csv



Input Layer $\in \mathbb{R}^1$

Hidden Layer $\in \mathbb{R}^3$

Output Layer $\in \mathbb{R}^1$